

Stochastic hierarchy induction for time series: Foundations and benchmark

This supplementary document contains additional figures that complement the main text of the paper titled: “Stochastic hierarchy induction for time series: Foundations and benchmark”. Each figure is labeled and referenced in the main text where appropriate.

SUPPLEMENTARY FIGURES, TABLES

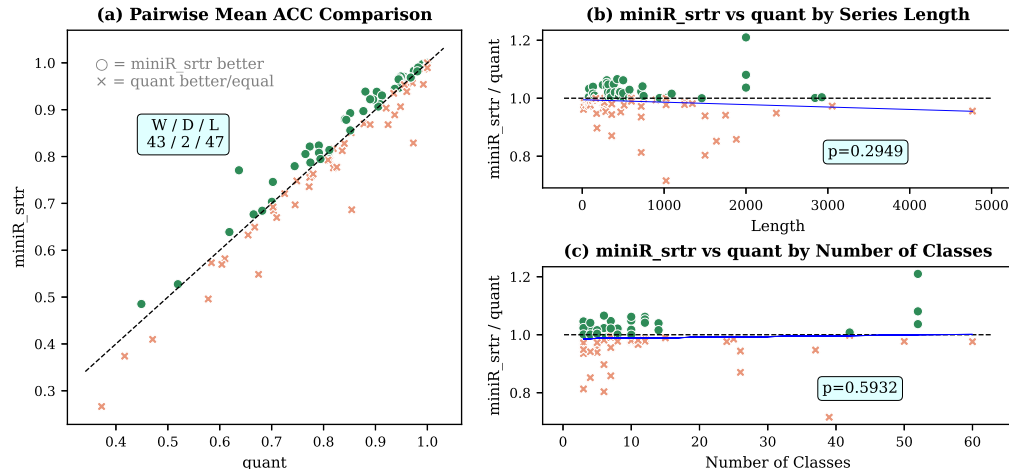


Fig. S1. Performance comparison of miniR_srttr versus quant: (a) mean accuracy values, and how miniR_srttr’s advantage correlates with (b) series length and (c) number of classes, quantified by Pearson correlation.

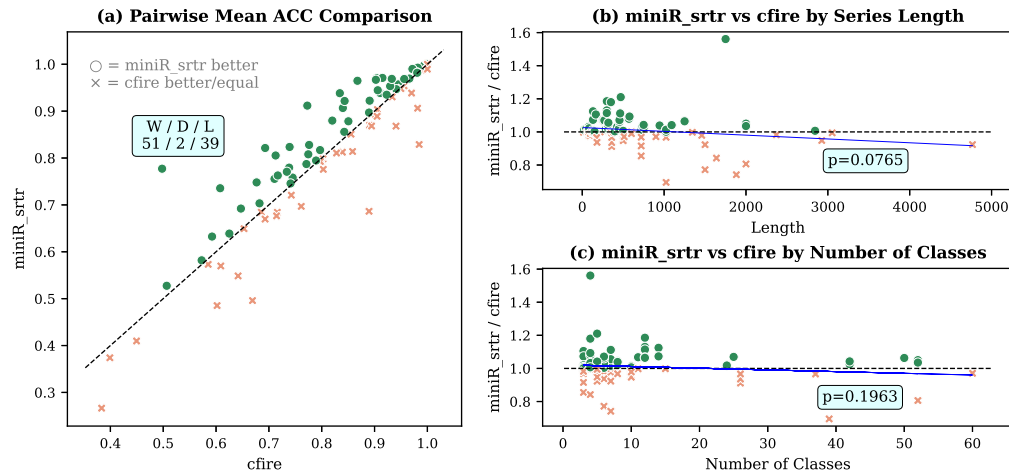


Fig. S2. Performance comparison of miniR_srttr versus cfire: (a) mean accuracy values, and how miniR_srttr’s advantage correlates with (b) series length and (c) number of classes, quantified by Pearson correlation.

Table S1. MCCV mean F1 scores and std values for selected FC and HiGEC schemes. Best mean score for each dataset is highlighted.

name	# class	miniR_srtr	miniR_potr	miniR_lsou	miniR	quant	cfire
ArrowHead	3	0.8798±0.0348	0.8768±0.0407	0.8781±0.0335	0.8825±0.0305	0.8411±0.0397	0.8200±0.0410
AsphaltPavementTypeUni_eq	3	0.8890±0.0082	0.8880±0.0066	0.8891±0.0083	0.8924±0.0082	0.9371±0.0081	0.9048±0.0082
BME	3	0.9927±0.0080	0.9916±0.0102	0.9918±0.0068	0.9944±0.0065	1.0000±0.0000	0.9818±0.0294
CBF	3	0.9970±0.0073	0.9959±0.0098	0.9946±0.0111	0.9971±0.0061	0.9906±0.0075	0.9933±0.0092
ChlorineConcentration	3	0.7480±0.0115	0.7480±0.0119	0.7474±0.0121	0.7549±0.0089	0.7484±0.0112	0.6768±0.0096
Covid3Month_disc	3	0.6388±0.0389	0.6295±0.0398	0.6361±0.0445	0.6273±0.0423	0.6186±0.0532	0.6251±0.0509
DistalPhalanxOutlineAgeGroup	3	0.7981±0.0221	0.7940±0.0240	0.8017±0.0233	0.7911±0.0255	0.8108±0.0242	0.8024±0.0310
InsectEPGRegularTrain	3	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
InsectEPGSmallTrain	3	0.9896±0.0126	0.9821±0.0238	0.9901±0.0129	0.9851±0.0256	1.0000±0.0000	1.0000±0.0000
LargeKitchenAppliances	3	0.8970±0.0162	0.8997±0.0148	0.8971±0.0158	0.9052±0.0143	0.8767±0.0200	0.8896±0.0144
Meat	3	0.9900±0.0140	0.9894±0.0146	0.9883±0.0144	0.9911±0.0127	0.9822±0.0166	0.9717±0.0220
MiddlePhalanxOutlineAgeGroup	3	0.6846±0.0351	0.6626±0.0467	0.6844±0.0342	0.6669±0.0369	0.7037±0.0287	0.6855±0.0329
ProximalPhalanxOutlineAgeGroup	3	0.8509±0.0198	0.8509±0.0194	0.8520±0.0198	0.8491±0.0196	0.8530±0.0189	0.8554±0.0191
RefrigerationDevices	3	0.6969±0.0396	0.6963±0.0404	0.6995±0.0400	0.7025±0.0460	0.7452±0.0387	0.7612±0.0394
ScreenType	3	0.5485±0.0241	0.5542±0.0289	0.5454±0.0210	0.5656±0.0266	0.6748±0.0353	0.6417±0.0339
SmallKitchenAppliances	3	0.8124±0.0136	0.8104±0.0133	0.8120±0.0144	0.8156±0.0138	0.8360±0.0138	0.8392±0.0158
SmoothSubspace	3	0.9540±0.0174	0.9540±0.0160	0.9540±0.0164	0.9644±0.0129	0.9922±0.0075	0.9558±0.0139
StarLightCurves	3	0.9818±0.0009	0.9817±0.0010	0.9819±0.0009	0.9815±0.0007	0.9807±0.0010	0.9807±0.0007
UMD	3	0.9826±0.0120	0.9836±0.0094	0.9831±0.0103	0.9854±0.0085	0.9738±0.0324	0.9806±0.0202
AsphaltObstaclesUni_eq	4	0.8783±0.0137	0.8779±0.0147	0.8739±0.0147	0.8710±0.0103	0.8437±0.0141	0.8505±0.0142
Car	4	0.9217±0.0273	0.9228±0.0290	0.9206±0.0288	0.9183±0.0314	0.8956±0.0487	0.8433±0.0484
CinCECGTorso	4	0.8289±0.0458	0.8299±0.0411	0.8279±0.0435	0.8597±0.0386	0.9730±0.0225	0.9844±0.0163
DiatomSizeReduction	4	0.9304±0.0332	0.9260±0.0338	0.9320±0.0329	0.9390±0.0339	0.9125±0.0321	0.9333±0.0307
EthanolLevel	4	0.7772±0.0275	0.7785±0.0281	0.7727±0.0288	0.6593±0.0233	0.8255±0.0270	0.4979±0.0272
FaceFour	4	0.9117±0.0375	0.9152±0.0398	0.9121±0.0379	0.9371±0.0363	0.9091±0.0460	0.7727±0.0625
OliveOil	4	0.9067±0.0278	0.9000±0.0310	0.9033±0.0388	0.9022±0.0430	0.9056±0.0334	0.8400±0.0540
Rock	4	0.7947±0.0615	0.7887±0.0595	0.8007±0.0625	0.8160±0.0504	0.7940±0.0549	0.7893±0.0653
Trace	4	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000	1.0000±0.0000
TwoPatterns	4	0.9999±0.0003	0.9994±0.0008	0.9990±0.0017	0.9945±0.0028	0.9998±0.0005	0.9988±0.0007
Beef	5	0.7356±0.0802	0.7456±0.0738	0.7300±0.0888	0.7678±0.0612	0.7722±0.0974	0.6078±0.0949
ECG5000	5	0.9469±0.0029	0.9463±0.0029	0.9474±0.0028	0.9468±0.0025	0.9456±0.0027	0.9410±0.0025
Haptics	5	0.5275±0.0214	0.5192±0.0234	0.5223±0.0216	0.5462±0.0161	0.5194±0.0197	0.5066±0.0240
MixedShapesRegularTrain	5	0.9686±0.0043	0.9671±0.0049	0.9691±0.0050	0.9705±0.0036	0.9679±0.0042	0.9562±0.0053
MixedShapesSmallTrain	5	0.9350±0.0101	0.9355±0.0102	0.9380±0.0082	0.9448±0.0072	0.9351±0.0109	0.9230±0.0121
PhoneHeartbeatSound	5	0.6493±0.0271	0.6487±0.0259	0.6460±0.0233	0.6526±0.0254	0.6674±0.0205	0.6529±0.0255
SemgHandSubjectCh2	5	0.8681±0.0183	0.8663±0.0183	0.8691±0.0136	0.8805±0.0112	0.9243±0.0193	0.9406±0.0145
Tools	5	0.8139±0.0262	0.8152±0.0258	0.8129±0.0284	0.8114±0.0269	0.8112±0.0290	0.8585±0.0237
Worms	5	0.7208±0.0560	0.7177±0.0547	0.7229±0.0480	0.7333±0.0620	0.7251±0.0418	0.7424±0.0491
Colposcopy	6	0.3739±0.0386	0.3677±0.0459	0.3779±0.0297	0.3686±0.0219	0.4168±0.0233	0.3990±0.0253
DistalPhalanxTW	6	0.7034±0.0259	0.7012±0.0336	0.7024±0.0249	0.6928±0.0249	0.7005±0.0176	0.6823±0.0217
GesturePebbleZ1_eq	6	0.9669±0.0176	0.9667±0.0198	0.9672±0.0176	0.9591±0.0199	0.9452±0.0193	0.9025±0.0217
GesturePebbleZ2_eq	6	0.9709±0.0206	0.9703±0.0204	0.9639±0.0297	0.9667±0.0219	0.9487±0.0253	0.9150±0.0271
MiddlePhalanxTW	6	0.5732±0.0281	0.5619±0.0256	0.5751±0.0304	0.5576±0.0232	0.5840±0.0200	0.5851±0.0244
OSULeaf	6	0.9386±0.0185	0.9486±0.0160	0.9485±0.0143	0.9599±0.0103	0.8806±0.0189	0.9106±0.0191
ProximalPhalanxTW	6	0.8169±0.0173	0.8143±0.0243	0.8172±0.0173	0.7979±0.0199	0.8187±0.0210	0.7969±0.0187
SemgHandMovementCh2	6	0.6864±0.0231	0.6880±0.0255	0.6659±0.0267	0.6836±0.0239	0.8543±0.0179	0.8896±0.0135
SyntheticControl	6	0.9869±0.0043	0.9856±0.0059	0.9871±0.0059	0.9888±0.0055	0.9933±0.0040	0.9901±0.0054
DodgerLoopDay_nmv	7	0.5818±0.0468	0.5797±0.0495	0.5680±0.0381	0.6039±0.0486	0.6100±0.0306	0.5727±0.0454
ElectricDevices	7	0.8698±0.0300	0.8720±0.0318	0.8641±0.0275	0.8744±0.0248	0.8778±0.0254	0.8904±0.0237
Fish	7	0.9648±0.0110	0.9644±0.0127	0.9659±0.0139	0.9750±0.0098	0.9446±0.0158	0.8676±0.0188
InlineSkate	7	0.4959±0.0290	0.4977±0.0328	0.4976±0.0329	0.4848±0.0257	0.5779±0.0320	0.6690±0.0325
KeplerLightCurves	7	0.9063±0.0135	0.9043±0.0131	0.9048±0.0126	0.9017±0.0114	0.9483±0.0106	0.9815±0.0057
Lightning7	7	0.7795±0.0553	0.7776±0.0611	0.7881±0.0522	0.7808±0.0540	0.7443±0.0460	0.7384±0.0452
Plane	7	0.9990±0.0029	0.9997±0.0017	0.9997±0.0017	1.0000±0.0000	1.0000±0.0000	0.9997±0.0017
Mallat	8	0.9386±0.0294	0.9381±0.0302	0.9294±0.0374	0.9524±0.0216	0.9603±0.0212	0.9701±0.0070

Table S2. Continuation of Table S2.

name	# class	miniR_srtr	miniR_potr	miniR_lsco	miniR	quant	cfire
UWaveGestureLibraryAll	8	0.9688±0.0034	0.9684±0.0035	0.9692±0.0034	0.9734±0.0032	0.9681±0.0043	0.9326±0.0053
UWaveGestureLibraryX	8	0.8558±0.0082	0.8587±0.0082	0.8490±0.0070	0.8568±0.0061	0.8515±0.0061	0.8427±0.0073
UWaveGestureLibraryY	8	0.7871±0.0077	0.7823±0.0101	0.7817±0.0069	0.7863±0.0063	0.7746±0.0067	0.7710±0.0067
UWaveGestureLibraryZ	8	0.8079±0.0059	0.8084±0.0069	0.8000±0.0059	0.8034±0.0061	0.7911±0.0066	0.7746±0.0060
ACSF1	10	0.8107±0.0412	0.8127±0.0434	0.8150±0.0358	0.8383±0.0416	0.8107±0.0411	0.8280±0.0394
AllGestureWiimoteX_eq	10	0.6843±0.0210	0.6871±0.0206	0.7040±0.0188	0.7307±0.0184	0.6818±0.0206	0.7159±0.0150
AllGestureWiimoteY_eq	10	0.7458±0.0181	0.7369±0.0192	0.7508±0.0166	0.7754±0.0190	0.7023±0.0199	0.7410±0.0159
AllGestureWiimoteZ_eq	10	0.6766±0.0218	0.6700±0.0226	0.6791±0.0212	0.7008±0.0199	0.6652±0.0229	0.7145±0.0188
MedicalImages	10	0.7922±0.0158	0.7855±0.0168	0.7826±0.0158	0.8027±0.0133	0.8117±0.0112	0.7997±0.0116
MelbournePedestrian_nmv	10	0.9570±0.0053	0.9565±0.0036	0.9581±0.0039	0.9615±0.0033	0.9712±0.0030	0.9451±0.0049
PickupGestureWiimoteZ_eq	10	0.7927±0.0625	0.7880±0.0437	0.7940±0.0498	0.8187±0.0342	0.8087±0.0443	0.7973±0.0317
ShakeGestureWiimoteZ_eq	10	0.8927±0.0352	0.8753±0.0428	0.8733±0.0370	0.9167±0.0351	0.8513±0.0431	0.8933±0.0275
InsectWingbeatSound	11	0.6325±0.0130	0.6316±0.0131	0.6330±0.0146	0.6545±0.0120	0.6546±0.0130	0.5924±0.0146
PLAID_eq	11	0.9034±0.0111	0.8977±0.0136	0.9026±0.0146	0.9125±0.0146	0.9206±0.0125	0.9054±0.0101
CricketX	12	0.8054±0.0179	0.8055±0.0181	0.8096±0.0159	0.8269±0.0161	0.7650±0.0168	0.7127±0.0196
CricketY	12	0.8214±0.0206	0.8247±0.0188	0.8234±0.0175	0.8397±0.0171	0.7734±0.0194	0.6929±0.0234
CricketZ	12	0.8236±0.0175	0.8257±0.0181	0.8270±0.0173	0.8468±0.0141	0.7909±0.0156	0.7392±0.0156
EOGHorizontalSignal	12	0.8279±0.0483	0.8244±0.0523	0.8249±0.0563	0.8433±0.0463	0.8400±0.0304	0.7762±0.0450
EOGVerticalSignal	12	0.7628±0.0465	0.7591±0.0483	0.7700±0.0511	0.7916±0.0521	0.7799±0.0480	0.7169±0.0483
FaceAll	14	0.9702±0.0366	0.9700±0.0313	0.9703±0.0283	0.9832±0.0329	0.9550±0.0350	0.9042±0.0139
FacesUCR	14	0.9383±0.0083	0.9372±0.0103	0.9323±0.0094	0.9701±0.0050	0.9025±0.0100	0.8343±0.0150
SwedishLeaf	15	0.9480±0.0121	0.9519±0.0076	0.9535±0.0078	0.9625±0.0044	0.9584±0.0071	0.9499±0.0086
Crop	24	0.7582±0.0040	0.7477±0.0041	0.7510±0.0033	0.7647±0.0025	0.7766±0.0025	0.7450±0.0025
WordSynonyms	25	0.6922±0.0231	0.6886±0.0199	0.6650±0.0112	0.7702±0.0135	0.7027±0.0131	0.6470±0.0117
GestureMidAirD1_eq	26	0.6697±0.0277	0.6938±0.0273	0.6813±0.0245	0.7185±0.0236	0.7097±0.0248	0.6928±0.0347
GestureMidAirD2_eq	26	0.5697±0.0386	0.5790±0.0396	0.5703±0.0333	0.5944±0.0347	0.6041±0.0343	0.6090±0.0315
GestureMidAirD3_eq	26	0.4097±0.0361	0.3915±0.0361	0.3933±0.0388	0.4318±0.0272	0.4708±0.0406	0.4495±0.0338
Adiac	37	0.7758±0.0214	0.7771±0.0225	0.7846±0.0195	0.8009±0.0160	0.8190±0.0161	0.8030±0.0146
Phoneme	39	0.2665±0.0139	0.2358±0.0154	0.2549±0.0103	0.2921±0.0092	0.3724±0.0124	0.3834±0.0139
NonInvasiveFetalECGThorax1	42	0.9446±0.0084	0.9467±0.0073	0.9489±0.0046	0.9528±0.0045	0.9372±0.0054	0.9060±0.0056
NonInvasiveFetalECGThorax2	42	0.9533±0.0052	0.9557±0.0056	0.9605±0.0047	0.9630±0.0042	0.9556±0.0041	0.9295±0.0051
FiftyWords	50	0.7555±0.0209	0.7580±0.0159	0.7384±0.0145	0.8247±0.0124	0.7734±0.0103	0.7106±0.0159
PigAirwayPressure	52	0.4853±0.0454	0.5383±0.0366	0.5663±0.0385	0.8601±0.0229	0.4490±0.0470	0.6019±0.0392
PigArtPressure	52	0.9223±0.0264	0.9210±0.0213	0.9261±0.0218	0.9601±0.0157	0.8897±0.0422	0.8907±0.0403
PigCVP	52	0.7707±0.0323	0.7941±0.0232	0.7936±0.0270	0.9183±0.0187	0.6370±0.0382	0.7340±0.0275
ShapesAll	60	0.8683±0.0137	0.8820±0.0140	0.8802±0.0154	0.9343±0.0114	0.8897±0.0116	0.8944±0.0129
# winners		9	3	5	38	21	16

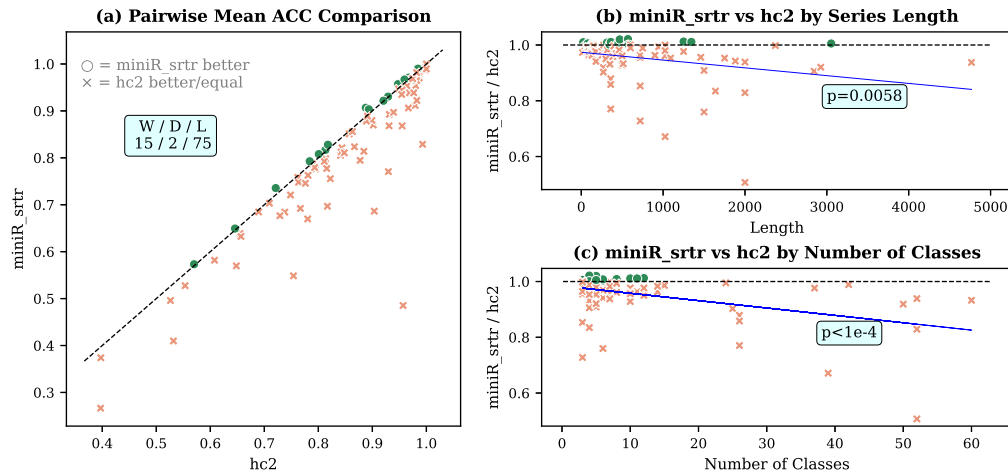


Fig. S3. Performance comparison of miniR_srtr versus hc2: (a) mean accuracy values, and how miniR_srtr's advantage correlates with (b) series length and (c) number of classes, quantified by Pearson correlation.

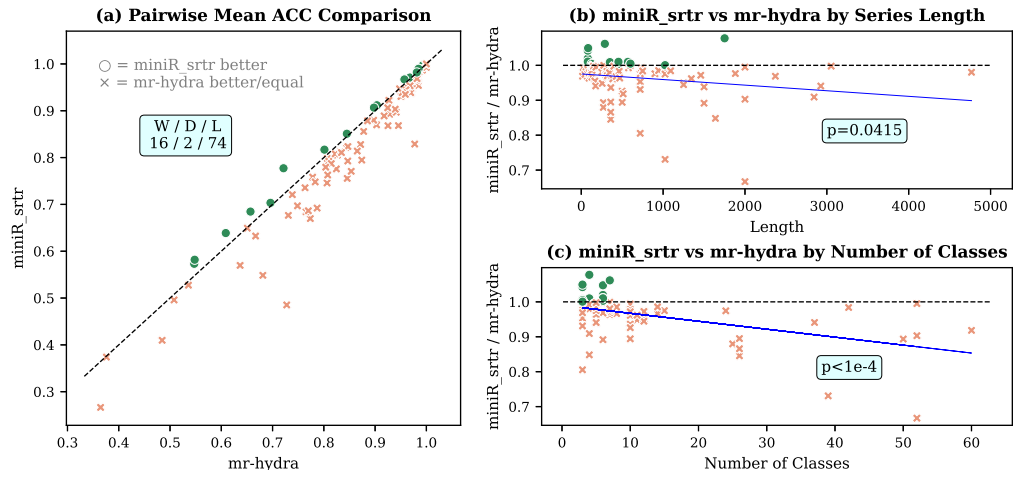


Fig. S4. Performance comparison of miniR_srtr versus mr-hydra: (a) mean accuracy values, and how miniR_srtr's advantage correlates with (b) series length and (c) number of classes, quantified by Pearson correlation.